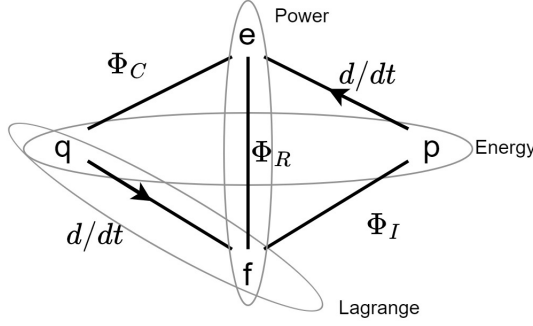


System Dynamics



Element (and Field) Types

Reactance(s)

- Efforts and flows
- Cannot store energy. Power conservative or power dissipative
- Hydraulic: $P_{12} = \Phi_R Q = \alpha Q |Q|$

$$\begin{array}{c} \text{Resistance form} \\ \vdots \frac{e_1}{f_1} \rightarrow R \leftarrow \frac{e_2}{f_2} \vdots \end{array} \quad \begin{array}{c} \text{Conductance form} \\ \vdots \frac{e_1}{f_1} \rightarrow R \leftarrow \frac{e_2}{f_2} \vdots \end{array}$$

$$\mathbf{e} = \Phi_R(\mathbf{f})$$

$$\mathbf{f} = \Phi_R^{-1}(\mathbf{e})$$

In conductance form:

$$f_2 = \frac{e_1 - e_2}{R} \quad f_1 = \frac{e_2 - e_1}{R}$$

Capacitance(s)

- Efforts and displacements
- $e = \Phi_C(q)$
- Hydraulic: $P = \rho g h = \rho g \frac{V}{A} = \frac{1}{C_f} V$

Integral Causality $\vdash \longrightarrow \mathbf{C}$

$$A \vdash \frac{e_A}{f_A} \longrightarrow \underbrace{\mathbf{C}}_{\mathbf{K}^{-1}} \frac{e_B}{f_B} \vdash B$$

$$\begin{bmatrix} e_A \\ e_B \end{bmatrix} = \mathbf{K} \begin{bmatrix} q_A \\ q_B \end{bmatrix}$$

$$E(\mathbf{q}) = \int_0^{\mathbf{q}} \Phi_C(\mathbf{q}) d\mathbf{q} \quad \frac{\partial e_i}{\partial q_j} = \frac{\partial e_j}{\partial q_i}$$

Inertance(s)

- Flows and momenta
- $e = \int \Phi_I(p) dp$ or $f = p/I$
- Hydraulic: $p_{p3} = \int (P_1 - P_2) dt \Rightarrow \dot{p}_{p3} = P_1 - P_2$
 – $Q = \frac{A}{\rho L} p_p = \frac{1}{I_f} p_p \Rightarrow$ Long thin pipes

– Pressure drop = d/dt (hydraulic momentum)

- Electrical: $p_L = Li$ (flux linkage)

Integral Causality $\longrightarrow \vdash \mathbf{I}$

$$A \xrightarrow{\frac{e_A}{f_A}} \underbrace{\mathbf{I}}_{\mathbf{I}^{-1}} \xrightarrow{\frac{e_B}{f_B}} B$$

$$\begin{bmatrix} f_A \\ f_B \end{bmatrix} = \mathbf{I}^{-1} \begin{bmatrix} p_A \\ p_B \end{bmatrix}$$

$$E(\mathbf{p}) = \int_0^{\mathbf{p}} \Phi_I(\mathbf{p}) d\mathbf{p} \quad \frac{\partial f_i}{\partial p_j} = \frac{\partial f_j}{\partial p_i}$$

IC-Fields

- Flows and momenta, efforts and displacements
- Two forms of energy storage

$$\begin{bmatrix} f_1 \\ \vdots \\ f_j \\ e_{j+1} \\ \vdots \\ e_n \end{bmatrix} = \mathbf{IC} \begin{bmatrix} p_1 \\ \vdots \\ p_j \\ q_{j+1} \\ \vdots \\ q_n \end{bmatrix} \quad \frac{\partial f_i}{\partial q_k} = \frac{\partial e_k}{\partial p_i}$$

$$E = \underbrace{\sum_{i=0}^j \int_0^{p_i} f_i dp_i}_{\text{I portion}} + \underbrace{\sum_{k=j+1}^n \int_0^{q_k} e_k dq_k}_{\text{C portion}}$$

Sources/Sinks

$$S_e \longrightarrow \vdash \quad S_f \vdash \longrightarrow$$

Transducers

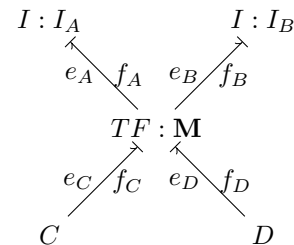
$$\longrightarrow \vdash \mathbf{GY} \vdash \longrightarrow \quad \text{OR} \quad \vdash \longrightarrow \mathbf{GY} \longrightarrow \vdash$$

$$\longrightarrow \vdash \mathbf{TF} \longrightarrow \vdash \quad \text{OR} \quad \vdash \longrightarrow \mathbf{TF} \vdash \longrightarrow$$

Multi-port Transformers

An example of a simplified system, where $\mathbf{M}$ comes from the original Bond Graph to satisfy the following two equations:

$$\begin{bmatrix} f_C \\ f_D \end{bmatrix} = \mathbf{M} \begin{bmatrix} f_A \\ f_B \end{bmatrix} \quad \begin{bmatrix} e_A \\ e_B \end{bmatrix} = \mathbf{M}^T \begin{bmatrix} e_C \\ e_D \end{bmatrix}$$



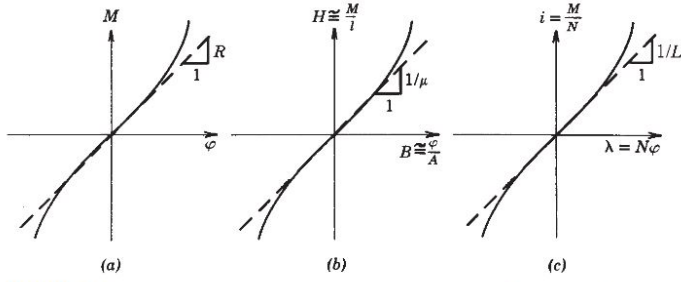
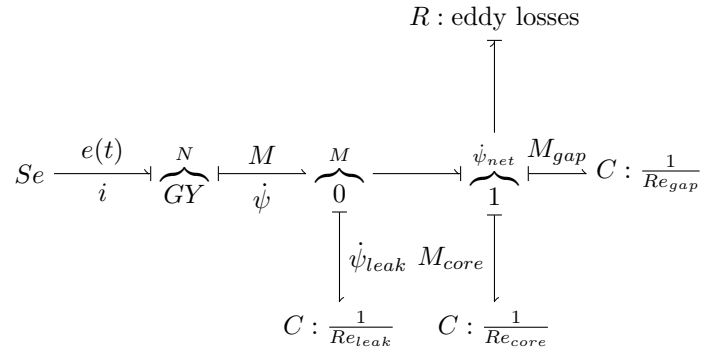


FIGURE 11.2. Core properties: (a) magnetomotive force versus flux; (b) field strength versus flux density; (c) current versus flux linkage.



Magnetic Domain

$$\lambda = N\psi \quad \dot{\lambda} = N\dot{\psi} \quad M = Ni$$

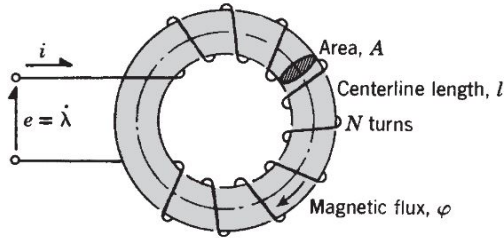
$$\frac{e = \dot{\lambda}}{i} \underbrace{GY}_N \frac{M}{\dot{\psi}}$$

$$\text{const. law of iron core: } M = Re\psi = \frac{L}{\mu A}\psi$$

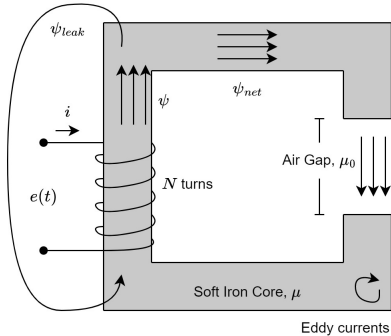
$$\text{OR} \quad H = \frac{1}{\mu}B \quad M = \frac{L}{\mu A}\psi$$

Electrical inductances are magnetic capacitances!

$$\frac{e}{i} \underbrace{GY}_N \frac{M}{\dot{\psi}} \quad C : \frac{1}{Re} \quad \text{OR} \quad \frac{e}{i} \underbrace{I : \frac{N^2}{Re}} = L$$

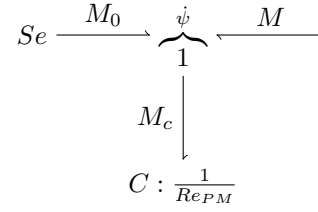


Example: Magnetic circuit and Bond Graph:



Permanent Magnets

$$M = M_c - M_0$$



Magnetic Forces

1. Lorentz Force

- DC motors, voice-coils, etc.
- Cannot store energy

$$\frac{e}{i} \underbrace{GY}_{k_t = \mathbf{L} \times \mathbf{B}} \frac{F_L}{\dot{x}}$$

2. Variable reluctance forces

$$e = e(x, \psi) \quad \Rightarrow \quad \text{C-field}$$

$$\frac{M_{gap}}{\dot{\psi}} \quad C \quad \frac{F}{\dot{x}}$$

$$\text{with: } M = M(x, \psi) \text{ and } F = F(x, \psi)$$

Thermodynamic Domain

$$\frac{du}{dt} = T\dot{s} - P\dot{v}$$

$$PV = mRT \rightarrow Pv = RT \quad \frac{\partial T}{\partial v} = \frac{\partial^2 u}{\partial v \partial s} = \frac{\partial(-P)}{\partial s}$$

$$\dot{E} = h_{up}\dot{m}_{up} = c_P T_{up}\dot{m}_{up} \quad \dot{m} = \frac{P_{in} - P_{out}}{R}$$

$$T = \frac{E}{mcv} \quad P = \frac{mRT}{V}$$

$$\frac{T}{\dot{S}} \quad C \quad \frac{P}{\dot{V}}$$

$$\begin{aligned}
\underbrace{T = T_0 e^{S/mC_V} \left(\frac{v}{v_0} \right)^{-R/C_V}}_{\text{integral thermal}} & \quad \underbrace{S = c_V \ln \left(\frac{T}{T_0} \right) + R \ln \left(\frac{v}{v_0} \right)}_{\text{derivative thermal}} \\
\underbrace{P = P_0 e^{S/mc_V} \left(\frac{v}{v_0} \right)^{-c_P/c_V}}_{\text{integral hydraulic}}
\end{aligned}$$

Special cases:

$$\begin{aligned}
& \frac{T}{\dot{S}} \rightarrow C \leftarrow \frac{P}{\dot{V}} \quad \underbrace{\text{const. pressure, state: } s}_{\dot{h} = \dot{u} + P\dot{v} = T\dot{s}} \\
& \frac{T}{\dot{S}} \rightarrow C \leftarrow \frac{P}{\dot{V}} \quad \underbrace{\text{const. temp, state: } v}_{\dot{f} = \dot{u} - T\dot{s} = -P\dot{v}} \\
& \frac{T}{\dot{S}} \rightarrow C \leftarrow \frac{P}{\dot{V}} \quad \underbrace{\text{const. temp \& press, no states}}_{\dot{\phi} = \dot{u} + P\dot{v} - T\dot{s} = 0} \\
& \frac{T}{\dot{S}} \rightarrow C \quad \underbrace{\text{isometric (thermal C), } \dot{V}=0}_{T = T_0 e^{S/mc_V}} \\
& C \leftarrow \frac{P}{\dot{V}} \quad \underbrace{\text{isentropic (pneumatic C), } \dot{S}=0}_{P = P_0 \left(\frac{v}{v_0} \right)^{-c_P/c_V}}
\end{aligned}$$

Heat Transfer:

$$\begin{aligned}
\dot{S}_2 - \dot{S}_1 &= \frac{H(T_1 - T_2)^2}{T_1 T_2} \geq 0 \\
\dot{S}_1 &= \frac{H(T_1 - T_2)}{T_1} \quad \dot{S}_2 = \frac{H(T_1 - T_2)}{T_2} \\
\frac{T_1}{\dot{S}_1} &\rightarrow R \leftarrow \frac{T_2}{\dot{S}_2}
\end{aligned}$$

- Conductive causality $\because \dot{S}_2 - \dot{S}_1 \geq 0$
- All electrical and mechanical R's are actually R-fields with the thermal domain. Causality on the thermal side is enforced, the other is not.

$$\begin{array}{cc}
\text{Electro-Thermal} & \text{Frictional Heating} \\
\frac{e(t)}{i} \rightarrow R \leftarrow \frac{T}{\dot{S}} & \frac{F}{\dot{x}} \rightarrow R \leftarrow \frac{T}{\dot{S}}
\end{array}$$

Reciprocity and Energy

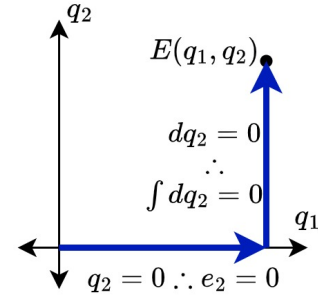
Example:

A C-field with half constitutive law $e_1 = e_1(q_1, q_2)$. The energy stored within a C-field is the following:

$$E = \int e_1 dq_1 + \int e_2 dq_2$$

Integrating cleverly in state space gives the total energy. Taking a derivative gets the other half of the constitutive law:

$$\frac{\partial E}{\partial q_2} = e_2(q_1, q_2)$$



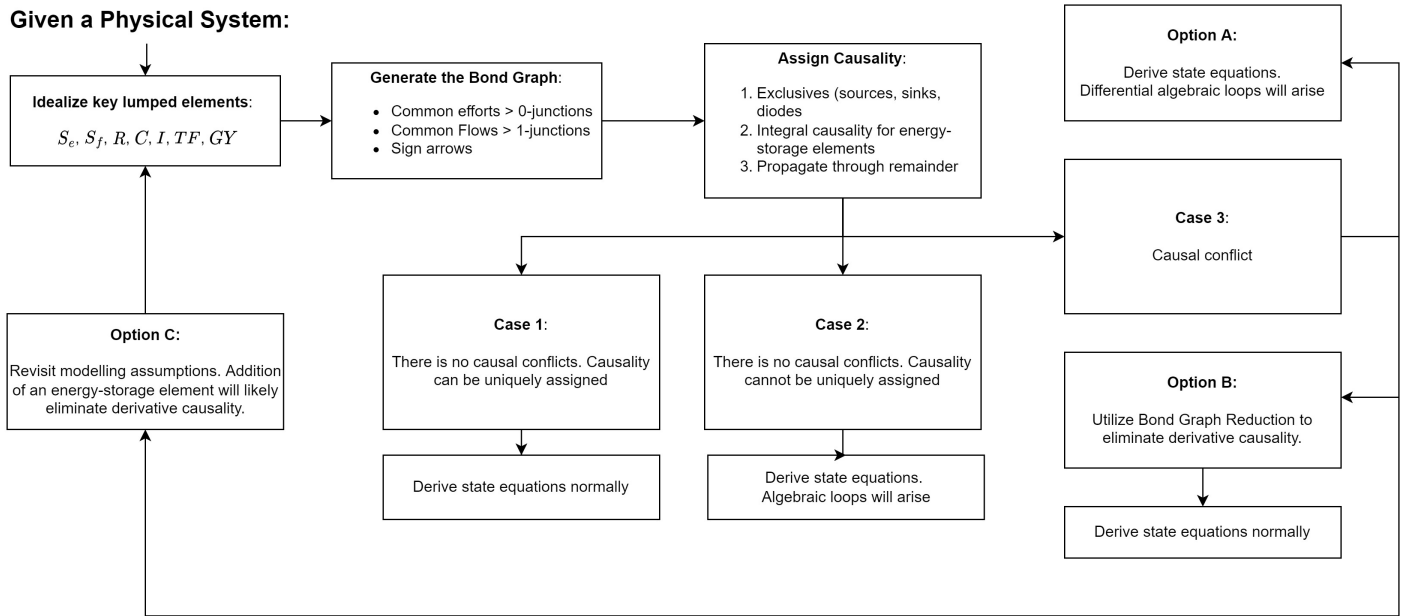
Linearization

An example C-field has constitutive law $V_c = V_c(q, x)$, with nominal values V_0, x_0, q_0 . A Taylor-series linearization then yields:

$$V_c \approx V_0 + \left. \frac{\partial V_c}{\partial q} \right|_{x_0, q_0} \Delta q + \left. \frac{\partial V_c}{\partial x} \right|_{x_0, q_0} \Delta x$$

Now, the states are Δq and Δx

Given a Physical System:



Domain	Effort	Flow
Mechanical		
• Translational	Force $[N]$	Velocity $[m/s]$
• Rotational	Torque $[Nm]$	Angular Velocity $[rad/s]$
Electrical	Voltage $[V]$	Current $[A]$
Magnetic	Magnetomotive Force $[A \cdot \text{turns}]$	Magnetic Flux Rate $[Weber/s = V]$
Fluidic		
• Incompressible	Pressure $[N/m^2]$	Volume Flow Rate $[m^3/s]$
• Compressible	Specific Enthalpy $[Nm/kg]$	Mass Flow Rate $[kg/s]$
Thermal	Temperature $[K]$	Entropy $[Nm/Ks]$
Chemical	Chemical Potential $[J/mol]$	Molar Flow Rate $[mass\ flow/Z]^*$